A maximal coupling of two laws on the positive integers: Y = f(X;U,U') with P[X != Y] = d_TV(mu,nu) and Y of law nu

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Editorial note. The lemma is attributed to Arratia in the source heading; the authors reproduce his proof at 1767-1793 and that reproduction is the proof carried here.

Editorial note. The supporting counting-measure identity 1715-1737 is included because the main proof invokes it at 1783; it is labelled as a supporting obligation and no separate count is assigned to it.

Editorial note. The map f_{mu,nu} is the same function the paper later uses to extract N_x (line 488); the context anchor 440-441 records that role without importing any part of the coupling theorem.

Editorial note. All displayed statements are taken from the fixed publisher source unchanged.

Allowed standard prerequisites (editorial).

- Total variation distance of two laws on the positive integers. d_TV(mu,nu) = sup_A |mu(A) - nu(A)|, and for laws supported on N it equals sum_i (mu(i)-nu(i))^+ = sum_i (mu(i)-nu(i))^-.. Supplies the quantity the constructed coupling attains exactly; the identity form is proved inside this card rather than assumed.
- Uniform randomisation on the unit square. Given X of law mu and two independent uniforms U, U' on (0,1), the two-branch map f_{mu,nu}(X;U,U') changes value exactly when U*mu(X) > nu(X).. Converts the total variation identity into an explicit coupling with Y measurable with respect to (X,U,U'); verifying the law of Y is the second obligation.

1 The coupling

The *total variation distance* is a metric between two probability distributions. Let μ and ν be two probability measures on $\mathbb{C}$. Then

$$\mathrm{d}_{\mathrm{TV}}(\mu,\nu) \coloneqq \sup |\mu(A) - \nu(A)|, \tag{1}$$

where the supremum is taken over all Lebesgue-measurable subsets of $\mathbb{C}$. For any real number a, let

$$a^+ \coloneqq \max\{a, 0\} \quad \text{and} \quad a^- \coloneqq \max\{-a, 0\}.$$

When μ and ν are supported on $\mathbb{Z}_{\geq 1}$, here are some alternative definitions of the total variation distance:

The total variation distance will be especially useful in this paper because of the following proposition. In [2, §3.8], Arratia proved that for any two random variables X and Y returning positive integers, we can always construct X' and Y' within the same probability space such that:

- X' and Y' have the same distribution as X and Y, respectively;
- Y' is a function of X', U, V where (U, V) is a point uniformly chosen in the unit square independent of X';
- $\mathbb{P}[X' \neq Y'] = \mathrm{d_{TV}}(X, Y)$.

Lemma 1.1 (Arratia, [2])**.** *Let μ and ν be two probability measures supported on $\mathbb{N}$, let X be a random variable with law μ, and let U and U' be two uniform random variables in $(0, 1)$ such that X, U, U' are independent. Let $Y \coloneqq f_{\mu,\nu}(X; U, U')$ with $f_{\mu,\nu}$ defined as above. Then $\mathbb{P}[X \neq Y] = \mathrm{d_{TV}}(\mu, \nu)$ and $\mathbb{P}[Y \in A] = \nu(A)$.*

Lemma 1.2. *Let μ and ν be two probability measure supported on $\mathbb{N}$. Then*

$$\mathrm{d_{TV}}(\mu, \nu) = \sum_{i \geq 1} (\mu(i) - \nu(i))^+ = \sum_{i \geq 1} (\mu(i) - \nu(i))^-.$$

Proof. Let $E \coloneqq \{i \in \mathbb{N} : \mu(i) > \nu(i)\}$ and let $\xi \coloneqq \mu - \nu$. Note that for any $B \subseteq \mathbb{N}$, we have $\xi(B \cap E) \geq 0 \geq \xi(B \cap E^c)$. Therefore, for any $A \subseteq \mathbb{N}$, we have

$$\xi(A) = \xi(E) + \xi(A \cap E^c) - \xi(A^c \cap E) \leq \xi(E)$$

and

$$\xi(A) = \xi(E^c) + \xi(A \cap E) - \xi(A^c \cap E^c) \geq \xi(E^c).$$

Therefore, $\mathrm{d_{TV}}(\mu, \nu) = \sup_{A \subseteq \mathbb{N}} |\xi(A)| = \max\{\xi(E), -\xi(E^c)\}$. Note that $\xi(E) = \sum_{i \geq 1} (\mu(i) - \nu(i))^+$, that $-\xi(E^c) = \sum_{i \geq 1} (\mu(i) - \nu(i))^-$ and that

$$\sum_{i \geq 1} (\mu(i) - \nu(i))^+ = \sum_{i \geq 1} \left[(\mu(i) - \nu(i)) + (\mu(i) - \nu(i))^- \right] = \sum_{i \geq 1} (\mu(i) - \nu(i))^-.$$

The lemma follows. □

We repeat his proof here. Let μ and ν be two probability measures supported on $\mathbb{N}$. Let $z_j \coloneqq \sum_{i \leq j} \frac{(\mu(i) - \nu(i))^-}{\mathrm{d_{TV}}(\mu, \nu)}$ (with $z_0 \coloneqq 0$). We consider the function $f_{\mu,\nu} \colon \mathbb{N} \times (0, 1)^2 \to \mathbb{N}$ defined as

$$f_{\mu,\nu}(m; a, b) \coloneqq \begin{cases} m & \text{if } a \cdot \mu(m) \leq \nu(m), \\ \sum_{i \geq 1} i \cdot \mathbb{1}_{z_{i-1} < b \leq z_i} & \text{otherwise.} \end{cases}$$

This is the function used in Section 1 for the extraction of N_x. Note that if $a, b \in (0,1)$ and $m \in \mathbb{N}$, then we have the equivalency

$$f_{\mu,\nu}(m; a, b) \neq m \quad \Longleftrightarrow \quad a \cdot \mu(m) > \nu(m). \tag{2}$$

Indeed, the direction "$\Rightarrow$" is obvious. To see the converse direction, note that if $a \cdot \mu(m) > \nu(m)$, then $(\mu(m) - \nu(m))^- = 0$, and hence the interval $(z_{m-1}, z_m]$ is empty.

Proof. Using (2), we find that

$$\{X \neq Y\} = \{U \cdot \mu(X) > \nu(X)\}. \tag{3}$$

Furthermore, we directly compute that

$$\mathbb{P}\big[U \cdot \mu(m) > \nu(m), X = m\big] = (\mu(m) - \nu(m))^+. \tag{4}$$

Therefore,

$$\mathbb{P}[X \neq Y] = \mathbb{P}[U \cdot \mu(X) > \nu(X)] = \sum_{m \geq 1} (\mu(m) - \nu(m))^+ = \mathrm{d_{TV}}(\mu, \nu) \tag{5}$$

with Lemma 1.2.

Next, we prove that $\mathbb{P}[Y = n] = \nu(n)$ for any $n \in \mathbb{N}$. Note that $Y = n$ if, and only if, one of two disjoint events happen: either we have $U \cdot \mu(n) \leq \nu(n)$ with $X = n$, or we have $U \cdot \mu(X) > \nu(X)$ with $z_{n-1} < U' \leq z_n$. Therefore, with (4) and (5), we have

$$\begin{aligned}\mathbb{P}[Y = n] &= \mathbb{P}\big[U \cdot \mu(n) \leq \nu(n), X = n\big] + \mathbb{P}\big[U \cdot \mu(X) > \nu(X)\big] \cdot \mathbb{P}\big[z_{n-1} < U' \leq z_n\big] \\ &= \Big(\mu(n) - \big(\mu(n) - \nu(n)\big)^+\Big) + \mathrm{d_{TV}}(\mu, \nu) \cdot \frac{(\mu(n) - \nu(n))^-}{\mathrm{d_{TV}}(\mu, \nu)} = \nu(n),\end{aligned}$$

where we used (3) and (5) to show that $\mathbb{P}\big[U \cdot \mu(X) > \nu(X)\big] = \mathrm{d_{TV}}(\mu, \nu)$. This concludes the proof of the lemma. □

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